Practical 11

Purva Tagde
23070521111
Batch B1

Example:

Index.html

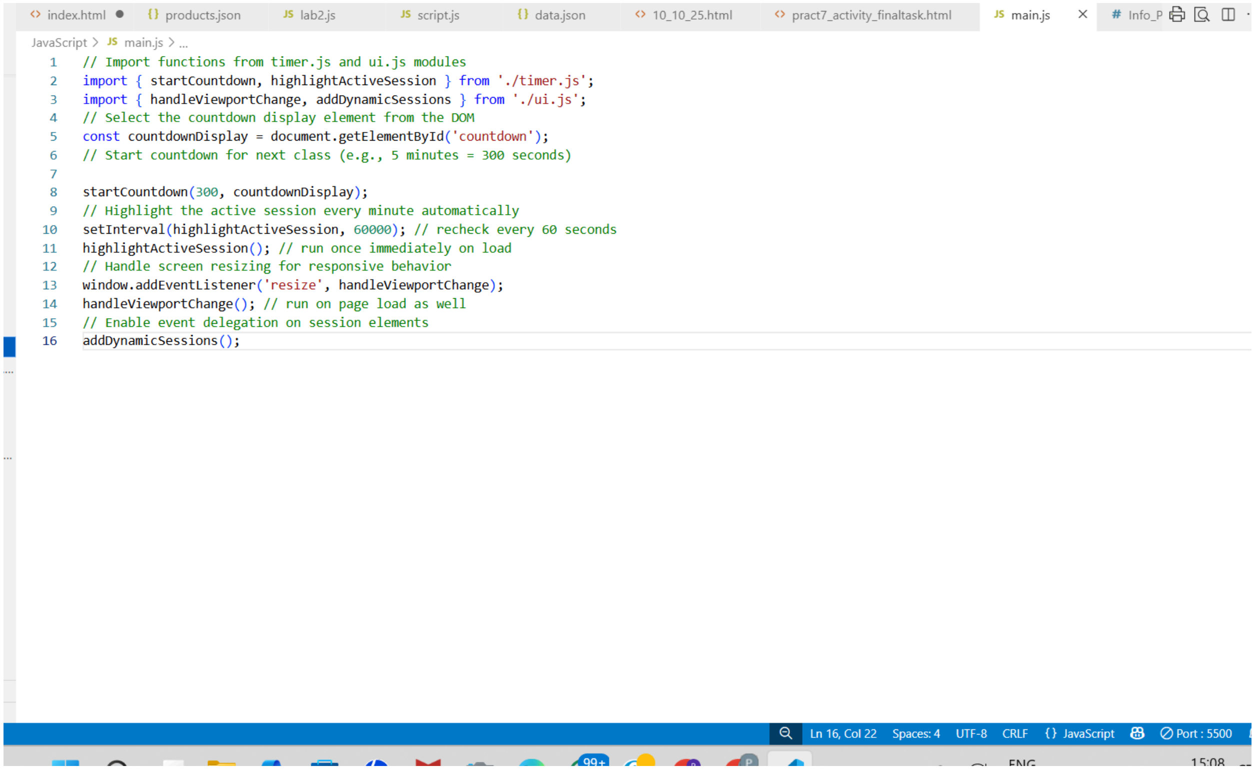
JavaScript > index.html > html > head > style > .active

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Timers, Dynamic UI & Modular Design</title>
<!-- Basic styling for the page -->
<style>
body { font-family: Arial; padding: 20px; }
/* Each class session block */
.session {
padding: 10px;
margin: 5px 0;
border: 1px solid #ccc;
border-radius: 5px;
}
/* Highlight active session */
.active {
background: #c8f7c5;
}
</style>
</head>
<body>
<h2>Class Scheduler (with Timers & Modules)</h2>
<!-- Container for all sessions -->
<div id="sessions">
<div class="session" data-time="09:00">HTML Basics - 9:00 AM</div>
<div class="session" data-time="10:00">CSS Design - 10:00 AM</div>
<div class="session" data-time="11:00">JavaScript Logic - 11:00 AM</div>
</div>
<!-- Display countdown timer -->
<p id="countdown"></p>
<!-- Main entry JavaScript file (module-based) -->
<script type="module" src="main.js"></script>
</body>
</html>
```

Ln 56, Col 21 Spaces: 2 UTF-8 CRLF {} HTML Port : 5500

ENG IN 15:08 05-11-2025

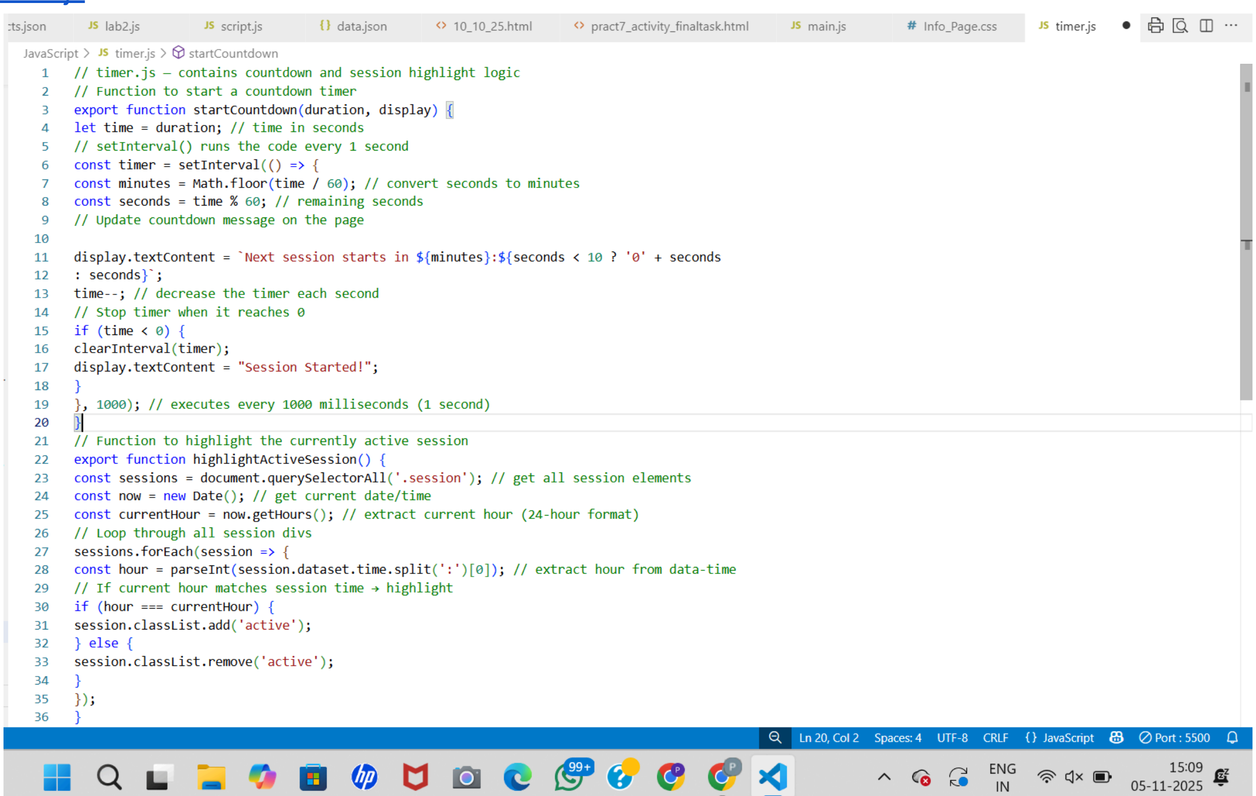
main.js



```javascript
// Import functions from timer.js and ui.js modules
import { startCountdown, highlightActiveSession } from './timer.js';
import { handleViewportChange, addDynamicSessions } from './ui.js';
// Select the countdown display element from the DOM
const countdownDisplay = document.getElementById('countdown');
// Start countdown for next class (e.g., 5 minutes = 300 seconds)

startCountdown(300, countdownDisplay);
// Highlight the active session every minute automatically
setInterval(highlightActiveSession, 60000); // recheck every 60 seconds
highlightActiveSession(); // run once immediately on load
// Handle screen resizing for responsive behavior
window.addEventListener('resize', handleViewportChange);
handleViewportChange(); // run on page load as well
// Enable event delegation on session elements
addDynamicSessions();
```

timer.js



```javascript
// timer.js — contains countdown and session highlight logic
// Function to start a countdown timer
export function startCountdown(duration, display) {
let time = duration; // time in seconds
// setInterval() runs the code every 1 second
const timer = setInterval(() => {
const minutes = Math.floor(time / 60); // convert seconds to minutes
const seconds = time % 60; // remaining seconds
// Update countdown message on the page

display.textContent = `Next session starts in ${minutes}:${seconds < 10 ? '0' + seconds
: seconds}`;
time--; // decrease the timer each second
// Stop timer when it reaches 0
if (time < 0) {
clearInterval(timer);
display.textContent = "Session Started!";
}
}, 1000); // executes every 1000 milliseconds (1 second)
}
// Function to highlight the currently active session
export function highlightActiveSession() {
const sessions = document.querySelectorAll('.session'); // get all session elements
const now = new Date(); // get current date/time
const currentHour = now.getHours(); // extract current hour (24-hour format)
// Loop through all session divs
sessions.forEach(session => {
const hour = parseInt(session.dataset.time.split(':')[0]); // extract hour from data-time
// If current hour matches session time → highlight
if (hour === currentHour) {
session.classList.add('active');
} else {
session.classList.remove('active');
}
});
}
```

ui.js

```
2.js  script.js  data.json  10_10_25.html  pract7_activity_finaltask.html  main.js  Info_Page.css  timer.js  ui.js
JavaScript > ui.js > addDynamicSessions
1  // ui.js – handles dynamic UI and responsive design behavior
2
3  // Function to change background color based on screen size
4  export function handleViewportChange() {
5  // If screen width < 600px, apply a light blue background
6  if (window.innerWidth < 600) {
7  document.body.style.background = '#f0f8ff';
8  } else {
9  // For larger screens, use white background
10 document.body.style.background = 'white';
11 }
12 }
13
14 // Function to add dynamic behavior (event delegation)
15 export function addDynamicSessions() {
16 const sessionsDiv = document.getElementById('sessions');
17 // Add event listener to parent container (event delegation)
18 sessionsDiv.addEventListener('click', (e) => {
19 // Check if clicked element has class "session"
20 if (e.target.classList.contains('session')) {
21 alert("You selected: " + e.target.textContent);
22 }
23 });
24 }
```

Ln 24, Col 2 Spaces: 4 UTF-8 CRLF {} JavaScript Port : 5500

ENG IN 15:09 05-11-2025

Output:

127.0.0.1:5500/JavaScript/index.html

-->

Class Scheduler (with Timers & Modules)

- HTML Basics - 9:00 AM
- CSS Design - 10:00 AM
- JavaScript Logic - 11:00 AM

Next session starts in 4:59

Camera

ENG IN 15:11 05-11-2025

Task:

index.html

JavaScript > index.html > html > head > style > #countdown

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Timers, Dynamic UI & Modular Design</title>
<style>
body {
  font-family: Arial, sans-serif;
  padding: 20px;
  transition: all 0.3s ease;
}
.session {
  padding: 10px;
  margin: 5px 0;
  border: 1px solid #ccc;
  border-radius: 5px;
  cursor: pointer;
}
.active {
  background-color: #c8f7c5;
}
#countdown {
  font-size: 18px;
  font-weight: bold;
  margin-top: 10px;
}
#countdown.red {
  color: red;
}
button {
  margin-top: 10px;
  padding: 8px 12px;
  background-color: #007bff;
  color: white;
  border: none;
  border-radius: 4px;
```


```
  border: none;
  border-radius: 4px;
  cursor: pointer;
}
button:hover {
  background-color: #0056b3;
}
@media (max-width: 500px) {
  body {
    font-size: 14px;
  }
}
</style>
</head>
<body>
<h2>Class Scheduler (with Timers & Modules)</h2>
<div id="sessions">
  <div class="session" data-time="09:00">HTML Basics - 9:00 AM</div>
  <div class="session" data-time="10:00">CSS Design - 10:00 AM</div>
  <div class="session" data-time="11:00">JavaScript Logic - 11:00 AM</div>
</div>
<button id="addBtn">Add Session</button>
<p id="countdown"></p>
<script type="module" src="main.js"></script>
</body>
</html>
```

helper.js

JavaScript > helper.js > ...

```
export function formatTime(seconds) {
  const minutes = Math.floor(seconds / 60);
  const sec = seconds % 60;
  return `${minutes}:${sec < 10 ? '0' + sec : sec}`;
}

export function logMessage(msg) {
  console.log(`[LOG] ${msg}`);
}
```

timer.js

JavaScript > timer.js > startCountdown > timer > setInterval() callback

```
import { formatTime, logMessage } from './helper.js';

export function startCountdown(duration, display) {
  let time = duration;
  const timer = setInterval(() => {
    display.textContent = `Next session starts in ${formatTime(time)}`;

    if (time < 60) {
      display.classList.add('red');
    } else {
      display.classList.remove('red');
    }
    time--;
    if (time < 0) {
      clearInterval(timer);
      display.textContent = "Session Started!";
      logMessage("Countdown finished.");
    }
  }, 1000);
}
export function highlightActiveSession() {
  const sessions = document.querySelectorAll('.session');
  const now = new Date();
  const currentHour = now.getHours();

  sessions.forEach(session => {
    const hour = parseInt(session.dataset.time.split(':')[0]);
    if (hour === currentHour) {
      session.classList.add('active');
    } else {
      session.classList.remove('active');
    }
  });
}
```

ui.js

JavaScript > ui.js > setupAddSessionButton

```
import { logMessage } from './helper.js';

export function handleViewportChange() {
  if (window.innerWidth < 600) {
    document.body.style.background = '#f0f8ff';
  } else {
    document.body.style.background = 'white';
  }
}

export function addDynamicSessions() {
  const sessionsDiv = document.getElementById('sessions');
  sessionsDiv.addEventListener('click', (e) => {
    if (e.target.classList.contains('session')) {
      alert("You selected: " + e.target.textContent);
      logMessage("Session clicked: " + e.target.textContent);
    }
  });
}

export function setupAddSessionButton() {
  const button = document.getElementById('addBtn');
  const sessionsDiv = document.getElementById('sessions');

  button.addEventListener('click', () => {
    const newDiv = document.createElement('div');
    newDiv.className = 'session';
    newDiv.dataset.time = '12:00';
    newDiv.textContent = 'New Added Session - 12:00 PM';
    sessionsDiv.appendChild(newDiv);
    alert('New session added!');
    logMessage("New session added dynamically.");
  });
}

```

Ln 22, Col 1 Spaces: 4 UTF-8 CRLF {} JavaScript Port : 5500

ENG IN 15:25 05-11-2025

main.js

JavaScript > main.js > ...

```
import { startCountdown, highlightActiveSession } from './timer.js';
import { handleViewportChange, addDynamicSessions, setupAddSessionButton } from './ui.js';

const countdownDisplay = document.getElementById('countdown');

startCountdown(300, countdownDisplay);

setInterval(highlightActiveSession, 60000);
highlightActiveSession();

window.addEventListener('resize', handleViewportChange);
handleViewportChange();

addDynamicSessions();

setupAddSessionButton();

```

Ln 1, Col 1 Spaces: 4 UTF-8 CRLF {} JavaScript Port : 5500

ENG IN 15:26 05-11-2025

OUTPUT:

Class Scheduler (with Timers & Modules)

HTML Basics - 9:00 AM

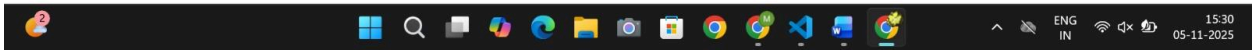
CSS Design - 10:00 AM

JavaScript Logic - 11:00 AM

Compiler Constuction - 15:31

Add Session

Next session starts in 4:39



Class Scheduler (with Timers & Modules)

HTML Basics - 9:00 AM

CSS Design - 10:00 AM

JavaScript Logic - 11:00 AM

Add Session

Session Started!

Class Scheduler (with Timers & Modules)

HTML Basics - 9:00 AM

CSS Design - 10:00 AM

JavaScript Logic - 11:00 AM

Add Session

Next session starts in 0:46